FUNCTIONAL ORGANIZATION OF RESPIRATORY SYSTEM

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DIKIOHS,DUHS.

LEARNING OBJECTIVES

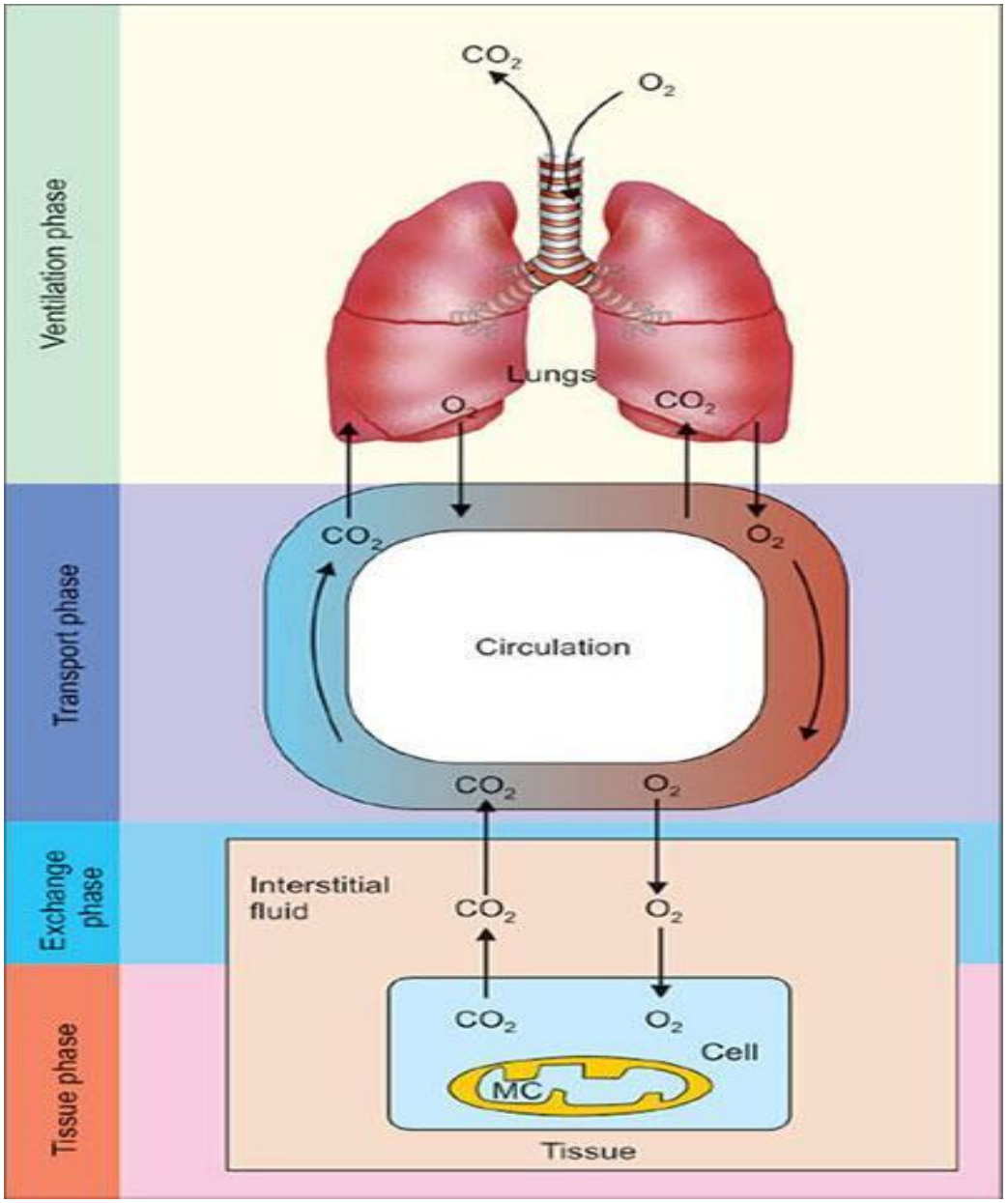
- At the end of lecture student should be able to:
- Define respiration
- Give functional organization of respiratory system
- Enlist various functions of respiration
- Distinguish between conducting & respiratory zone.
- Discuss the Non respiratory functions of the respiratory tract.

DEFINITION

At organism level, respiration is the process of gas exchange—the release of carbon dioxide and the uptake of oxygen that occurs between RBCs and alveoli.

At cellular level, respiration is the release of energy from the breakdown of food in the presence of oxygen.

STAGES OF RESPIRATION



VENTILATION/BREATHING

Breathing/Ventilation - refers to movement of air; in and out of lungs.

- consists of two cyclic phases:
  inhalation, also called **inspiration**
  exhalation, also called **expiration**
  Inhalation draws gases into the lungs
  Exhalation forces gases out of the lungs.

Respiration in three places

- **External respiration**
  - Exchange of gases between atmosphere and blood tissue
  - Occurs in alveoli of lung tissue
- **Internal respiration**
  - Exchange of gases between blood and cells of the body.
- **Cellular respiration**
  - Utilization of oxygen to create ATP through oxidative phosphorylation.

Cells Types of the Alveoli

- Type I alveolar cells (type 1 Pneumocytes)

  simple squamous cells forming alveolar epithelium where gas exchange occurs

- Type II alveolar cells (type 2 Pneumocytes)

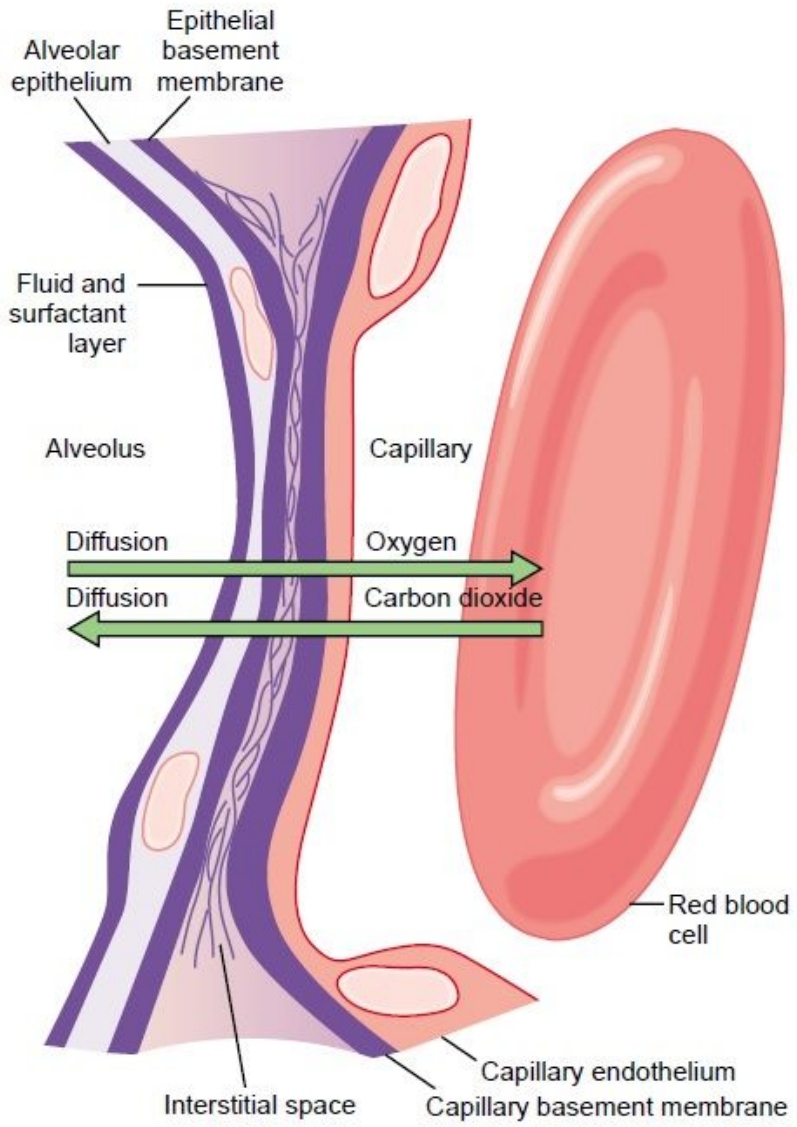
  secrete alveolar fluid containing surfactant

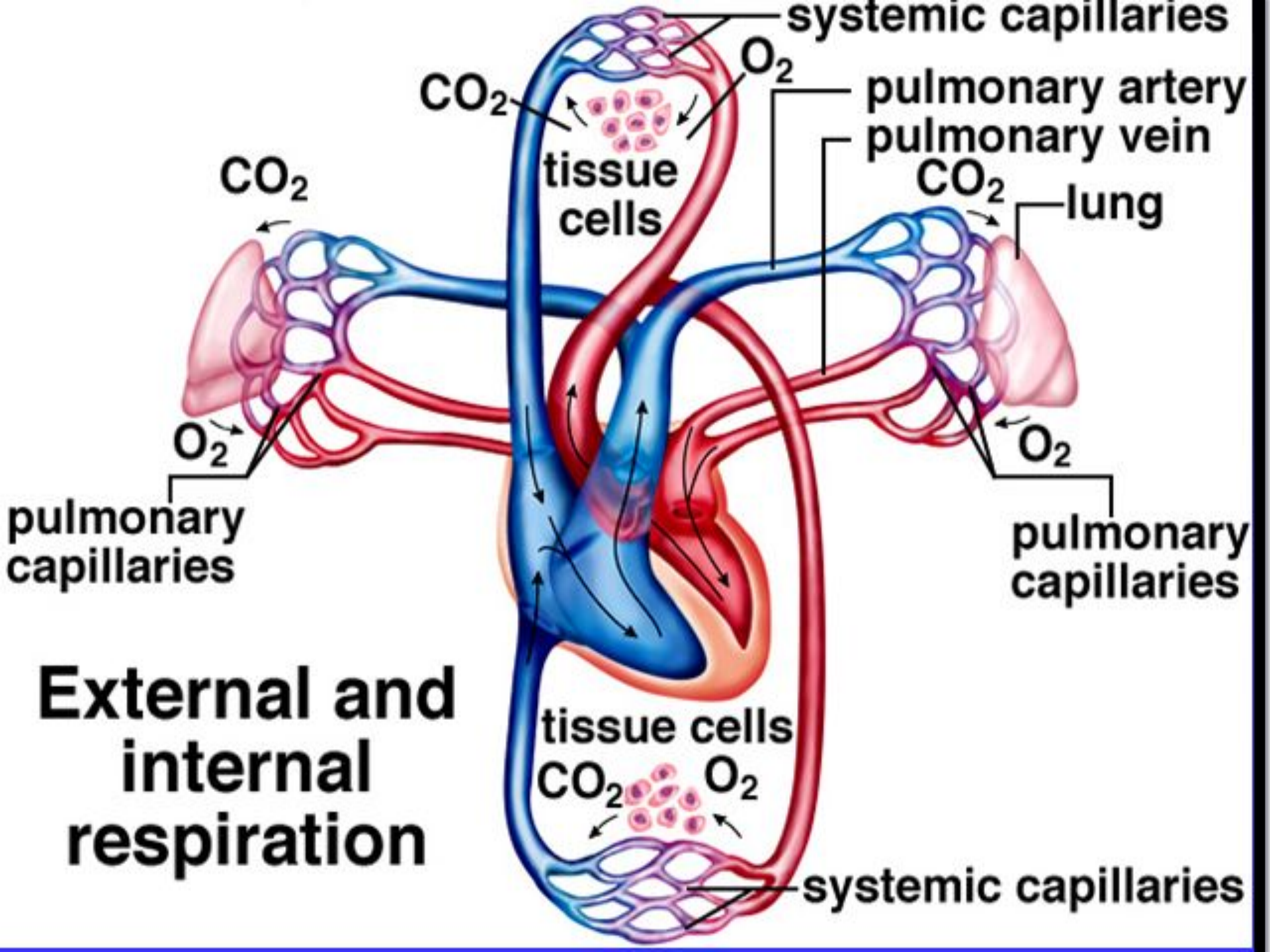
- Alveolar Macrophages

  wandering macrophages remove debris.

Respiratory Membrane

- Respiratory membrane = 0.3-0.5 µm thick and an area of 50 – 100 sq. meters
- Exchange of gas from alveoli to blood
- 6 Layers of membrane to cross
  1. layer of fluid and surfactant
  2. alveolar epithelial wall of type I cells
  3. alveolar epithelial basement membrane
  4. interstitial space
  5. capillary basement membrane
  6. endothelial cells of capillary




systemic capillaries
O_2
CO_2
pulmonary artery
pulmonary vein
tissue cells
CO_2
CO_2
lung
O_2
O_2
pulmonary capillaries
pulmonary capillaries
External and internal respiration
tissue cells
CO_2
O_2
systemic capillaries

Organization of respiratory system

- **Two methods of organization in use**
  - Organized according to **location** of structures
  - Organized according to **function** of structures

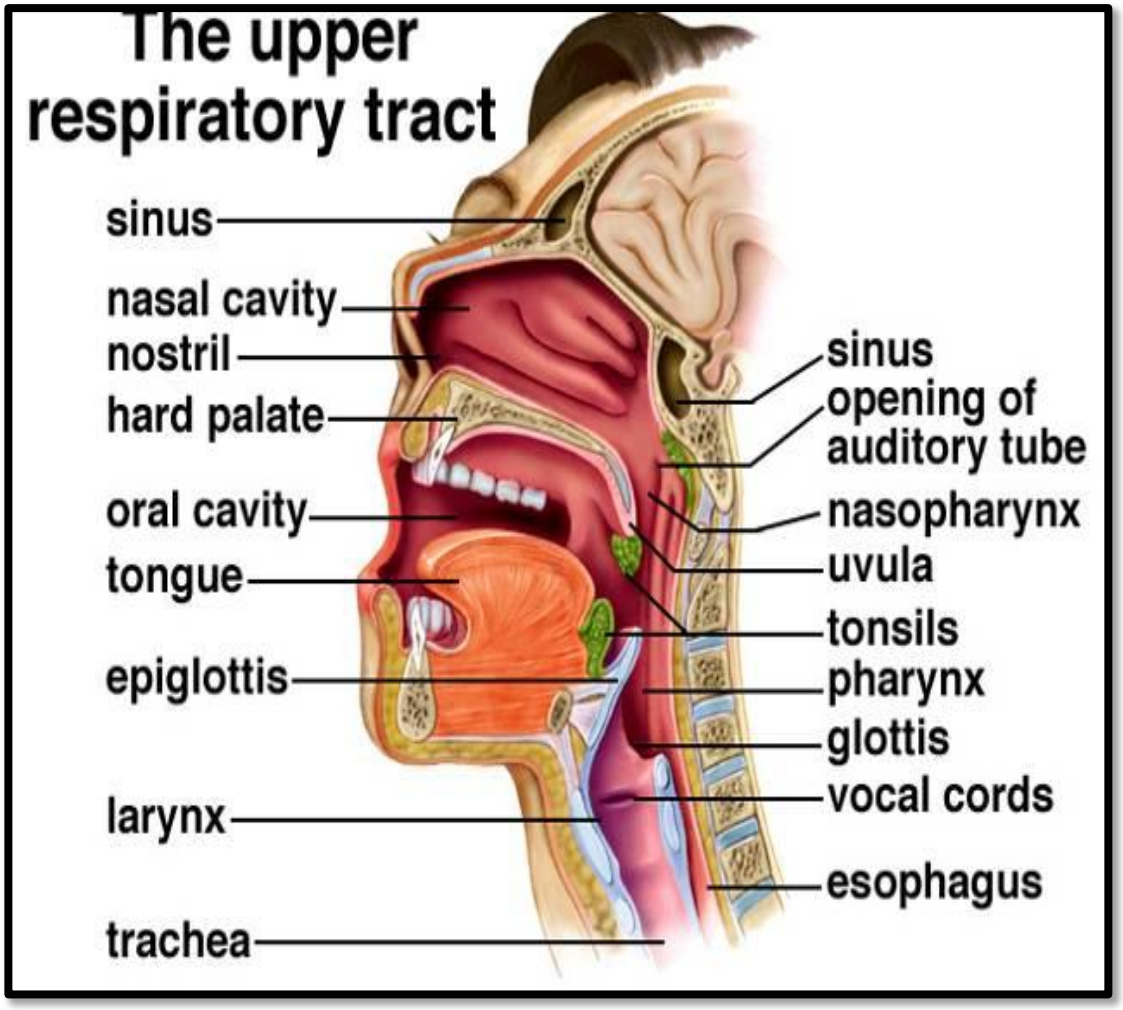
STRUCTURAL ORGANIZATION

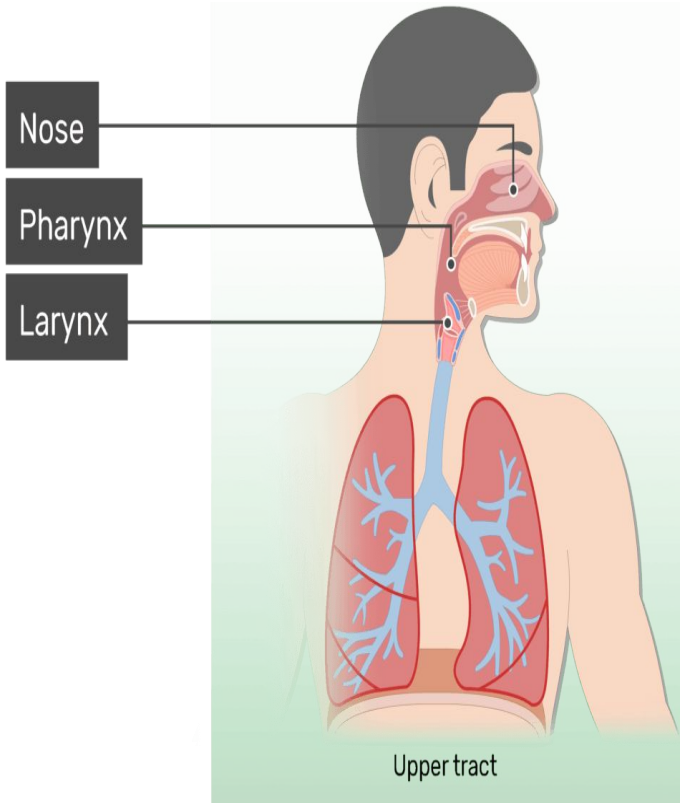
Generally, respiratory tract is divided into two parts:

1. **Upper respiratory tract** that includes all the structures from nose up to vocal cords; vocal cords are the folds of mucous membrane within larynx that vibrates to produce the voice.
2. **Lower respiratory tract**, which includes trachea, bronchi and lungs.

Upper respiratory Tract

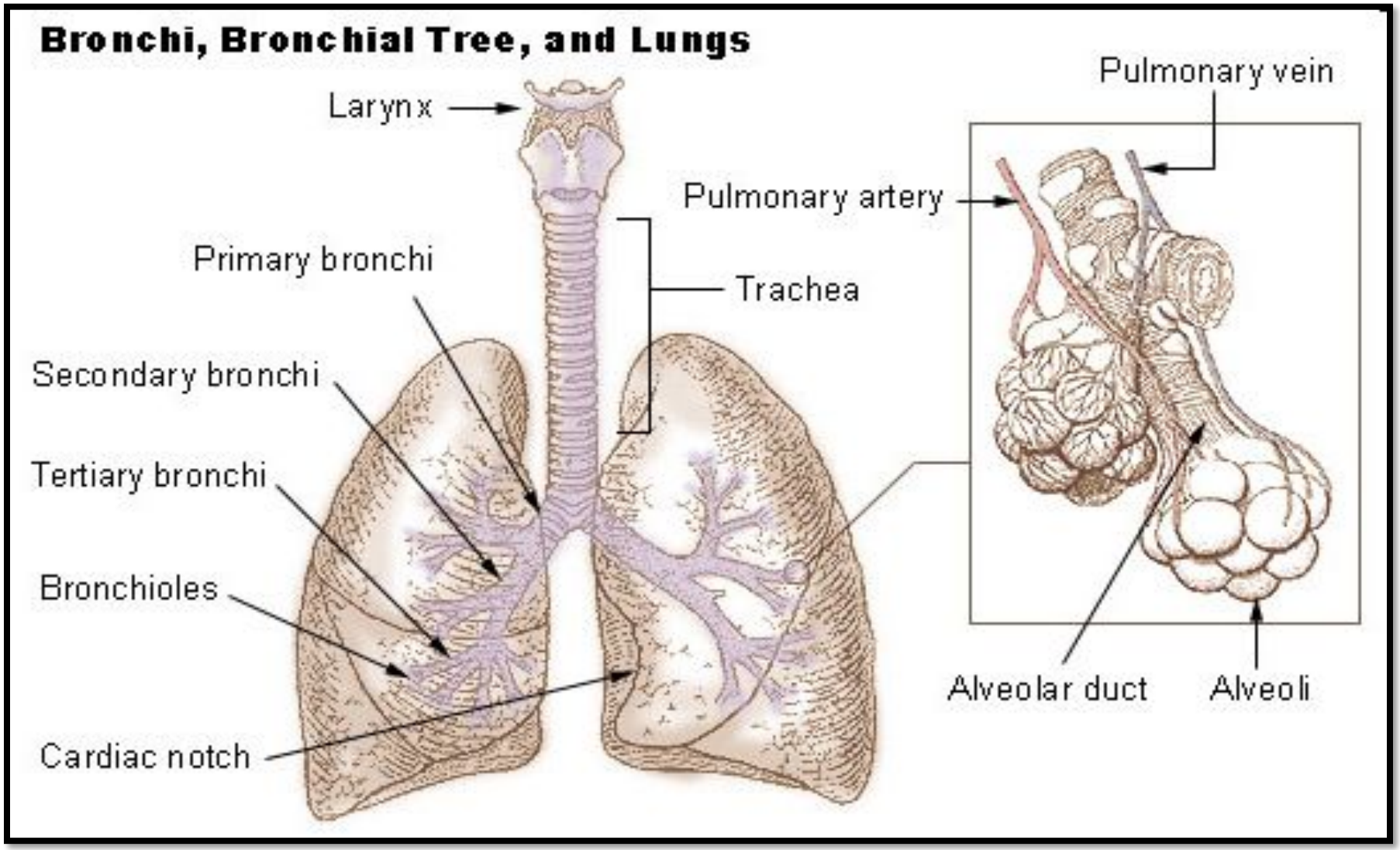
extends from the upper nares to the vocal cords.





Lower Respiratory Tract

Extends from the vocal cords to the alveoli

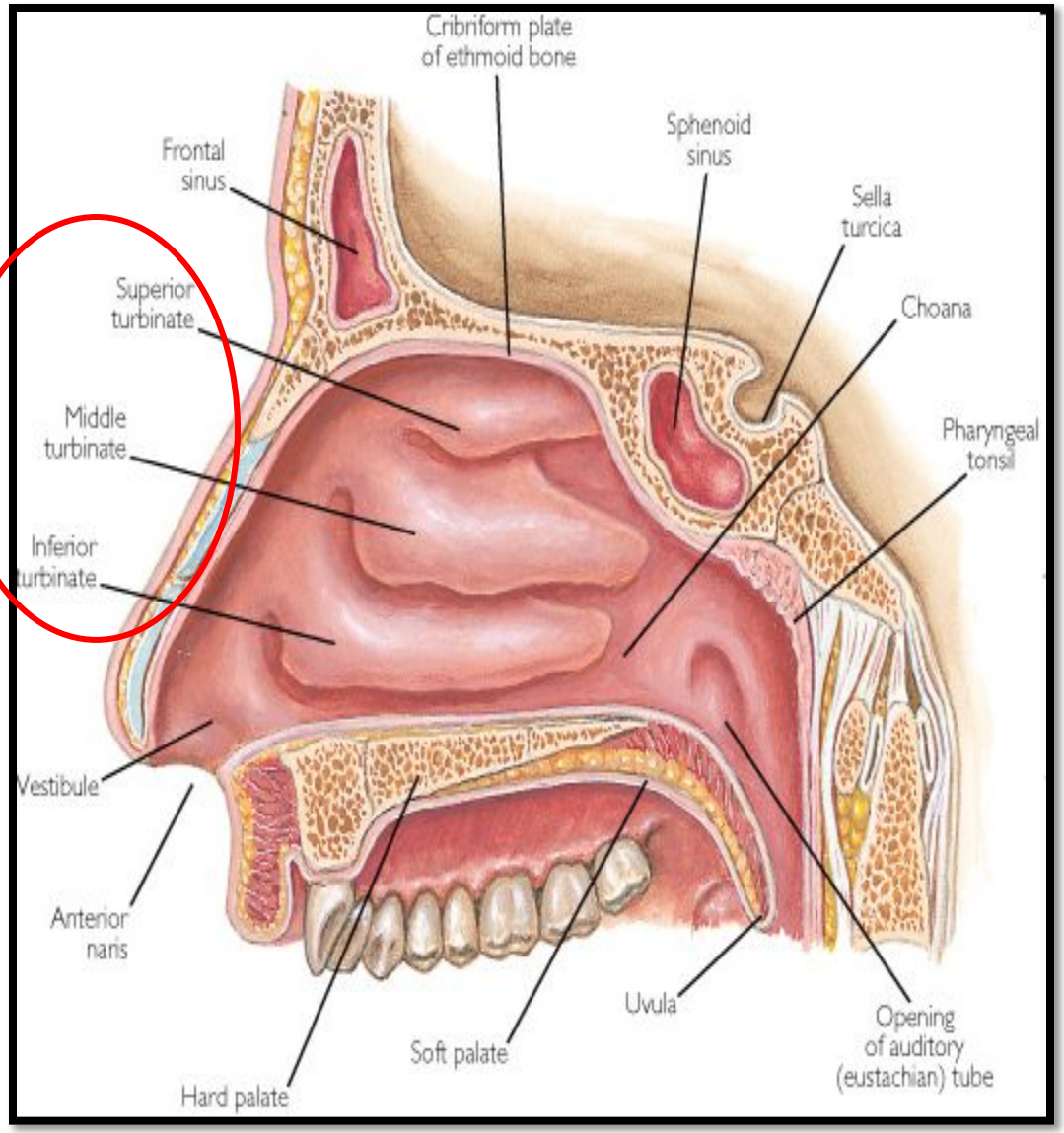


FUNCTIONAL CLASSIFICATION

- The respiratory system includes the lungs and a series of airways that connect the lungs to the external environment. The structures of the respiratory system are subdivided into
- **A conducting zone** (or conducting airways), which brings air into and out of the lungs,
- **A respiratory zone** lined with alveoli, where gas exchange occurs. The functions of the conducting and respiratory zones differ, and the structures lining them also differ.

Nasal Cavity

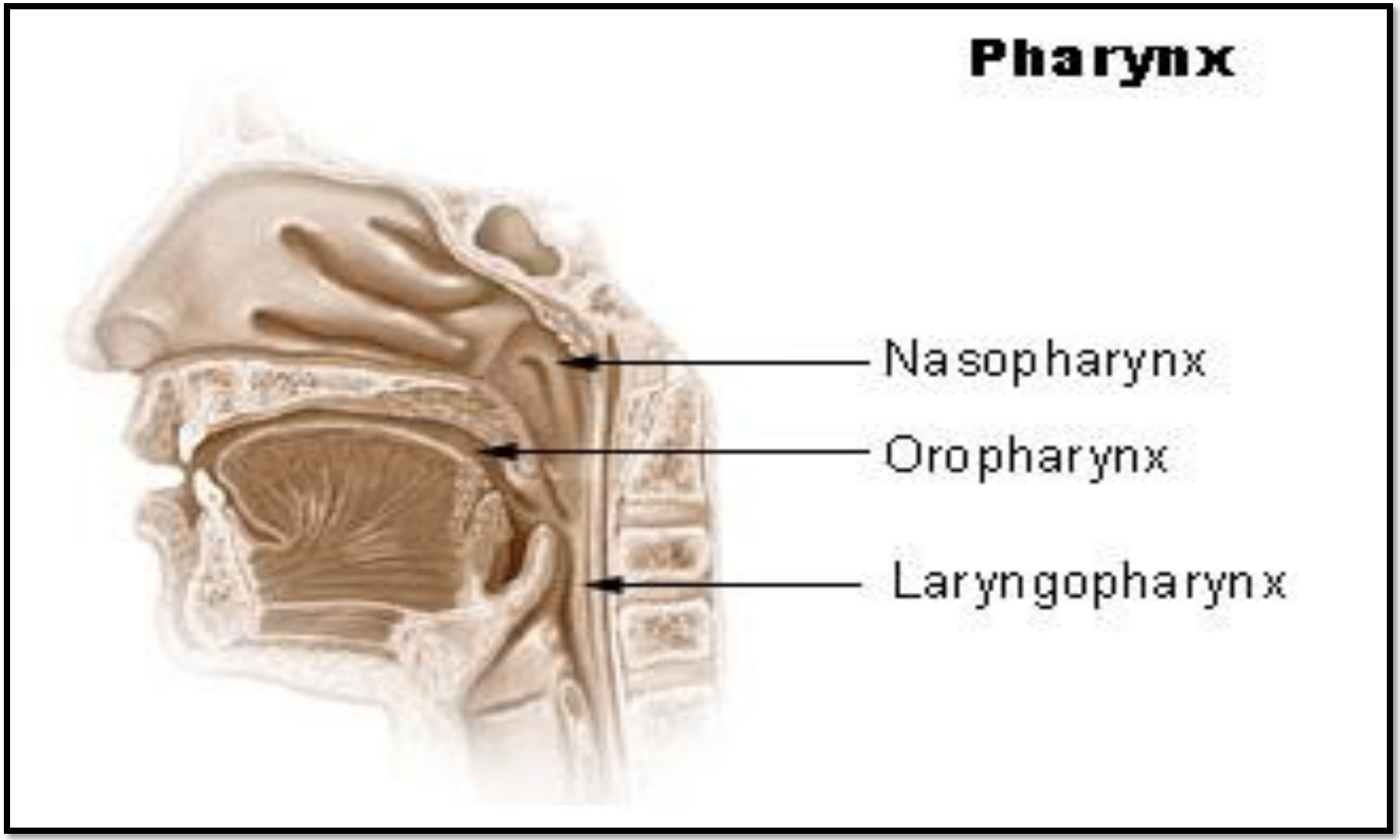
- **Nares**
  External opening
- **Septum**
  Divides the cavity
- **Conchae/ turbinates**
  Create channels for air to move through
  - Causes the air to swirl
  - Increase surface area and increases contact between air and mucosa.



Functions of Nasal Cavity

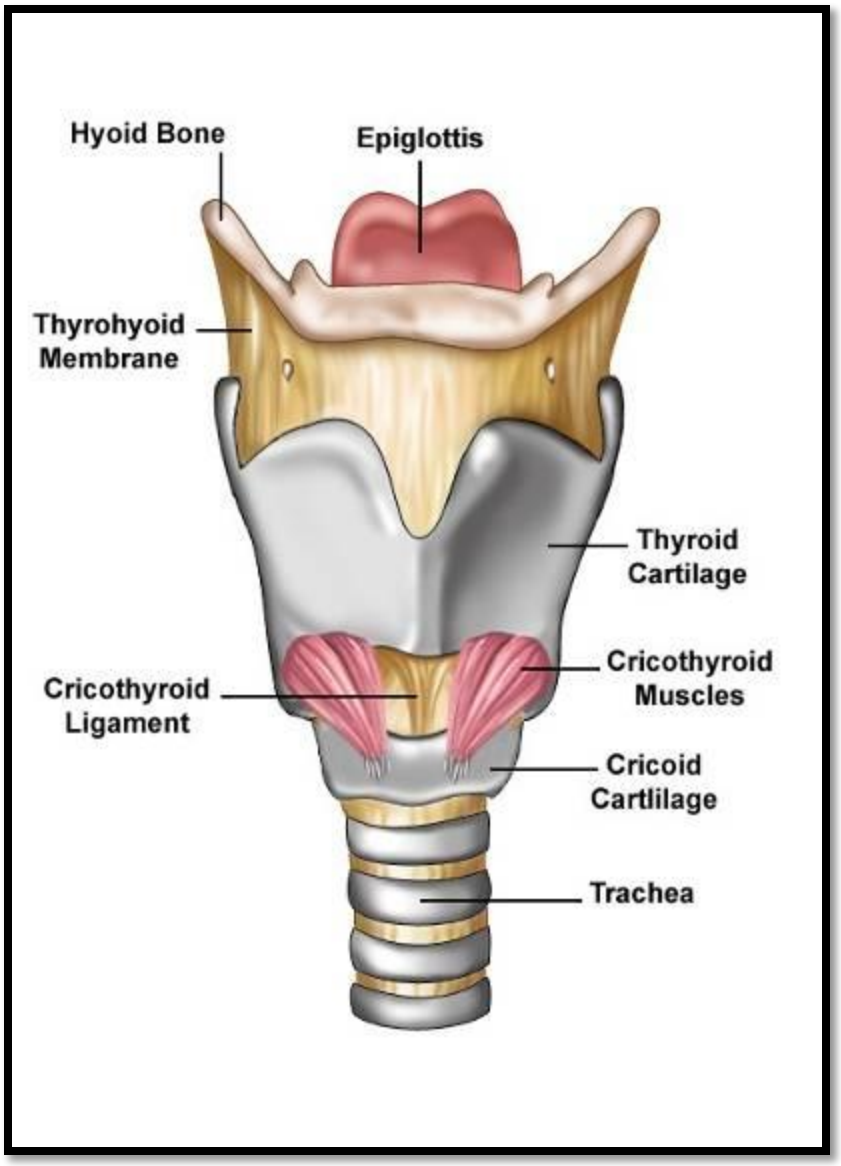
- **Filters air**
- **hair** (vibrissae) in the passage trap particles
- **Warms air**
- **Moisturizes air**
- **Olfaction**

Pharynx



Larynx

- Maintains patent airway
- Epiglottis **protects trachea** from food particles by routing food into esophagus.
- **Produces sound**



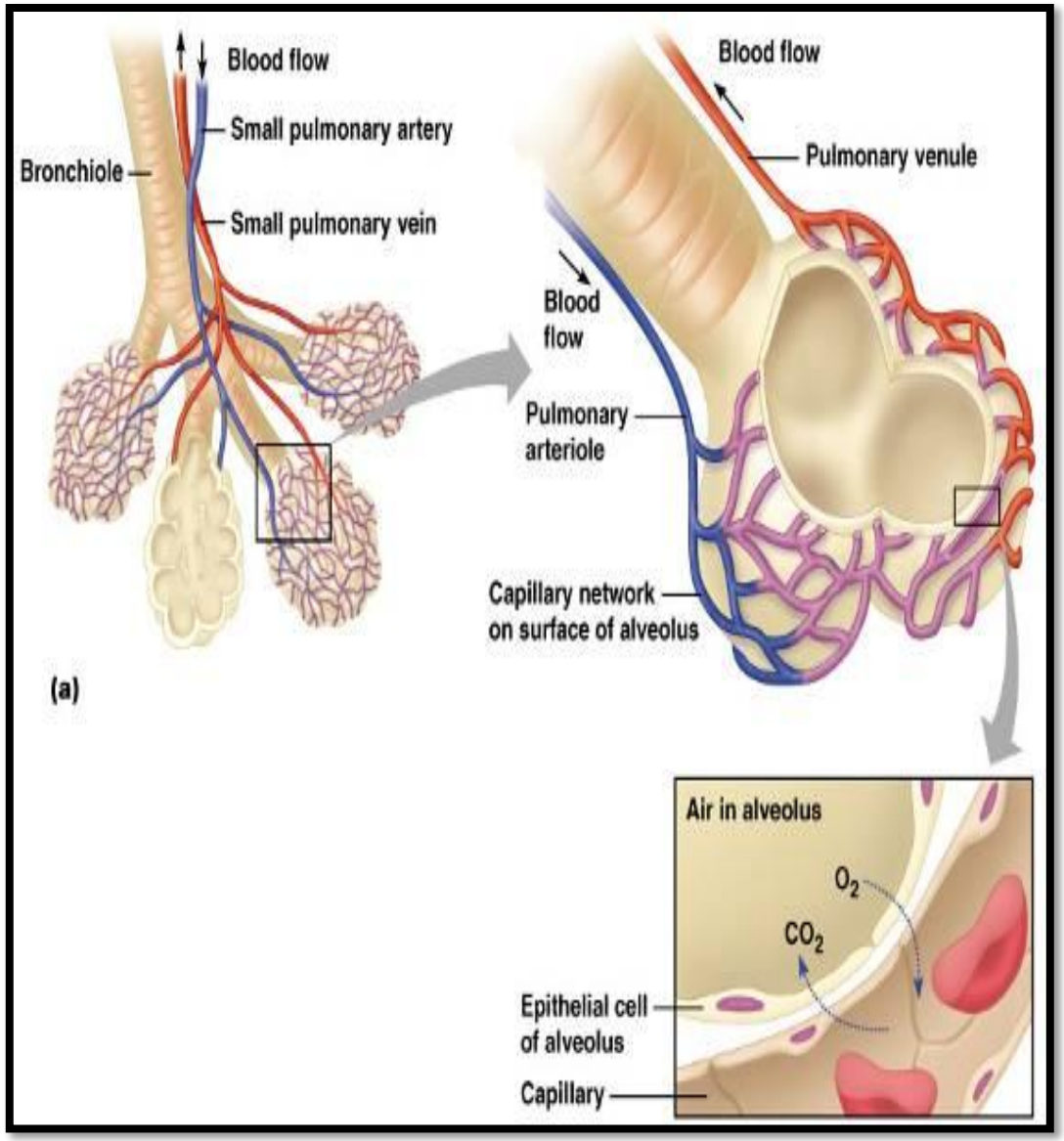
Functional Classification

- **Conducting zone**

Everything not involved in gas exchange.

- **Gas exchange or Respiratory zone**

Respiratory bronchioles and alveoli.



CONDUCTING ZONE

- The conducting zone includes the nose, nasopharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles.
- These structures function to bring air into and out of the respiratory zone for gas exchange and to warm, humidify, and filter the air before it reaches the critical gas exchange region.
- The walls of the conducting airways contain smooth muscle. This smooth muscle has both sympathetic and parasympathetic innervation, which have opposite effects on airway diameter:

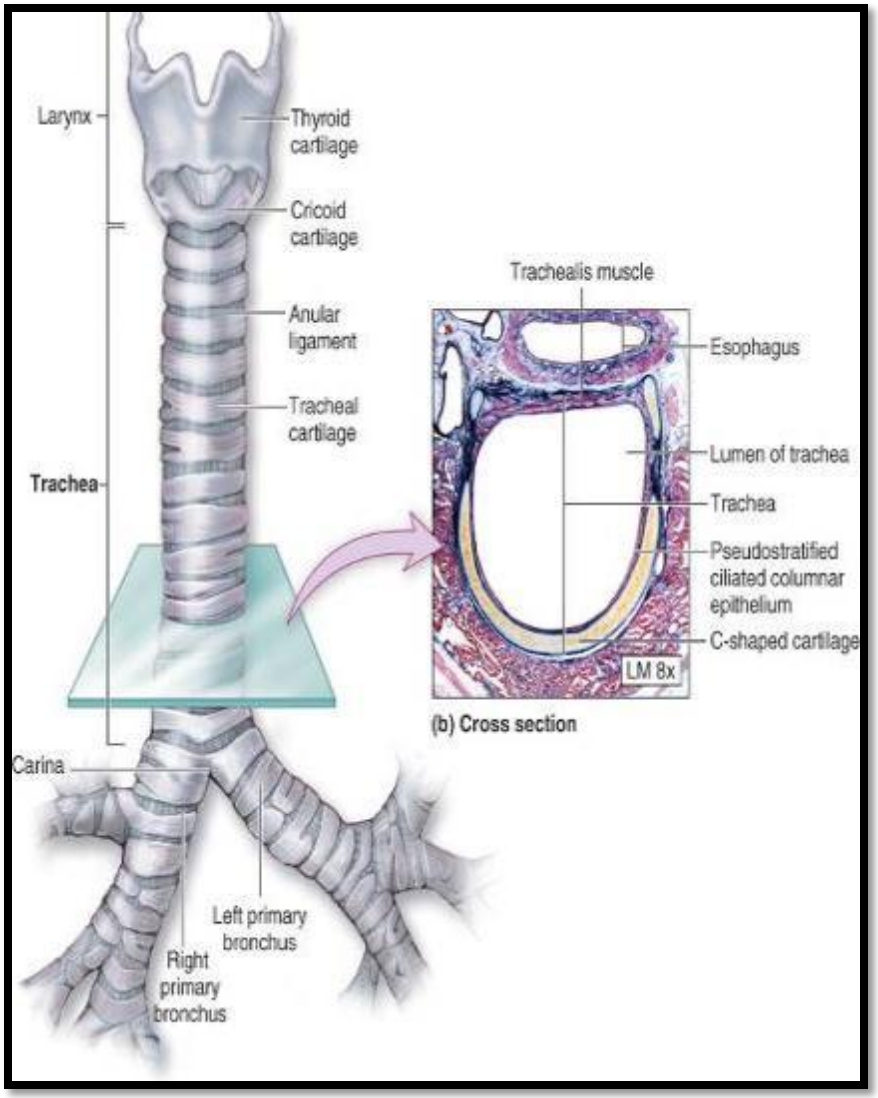
  (1) Sympathetic adrenergic neurons activate b2 receptors on bronchial smooth muscle, which leads to relaxation and dilation of the airways.

  (2) Parasympathetic cholinergic neurons activate muscarinic receptors, which leads to contraction and constriction of the airways.

- **The trachea** is the main conducting airway. The trachea divides into two bronchi, one leading into each lung, which divide into two smaller bronchi, which divide again.
- Ultimately, there are 23 such divisions into increasingly smaller airways. The conducting airways are lined with mucus-secreting and ciliated cells that function to remove inhaled particles. Although large particles usually are filtered out in the nose, small particles may enter the airways, where they are captured by mucus, which is then swept upward by the rhythmic beating of the cilia.

Trachea and Bronchi

- Extends from larynx into mediastinum.
- Carina- ridge marks bifurcation of primary bronchi
- C-shaped hyaline cartilage keep passage patent
- Posterior of trachea is made of trachealis muscle.



Primary Bronchi

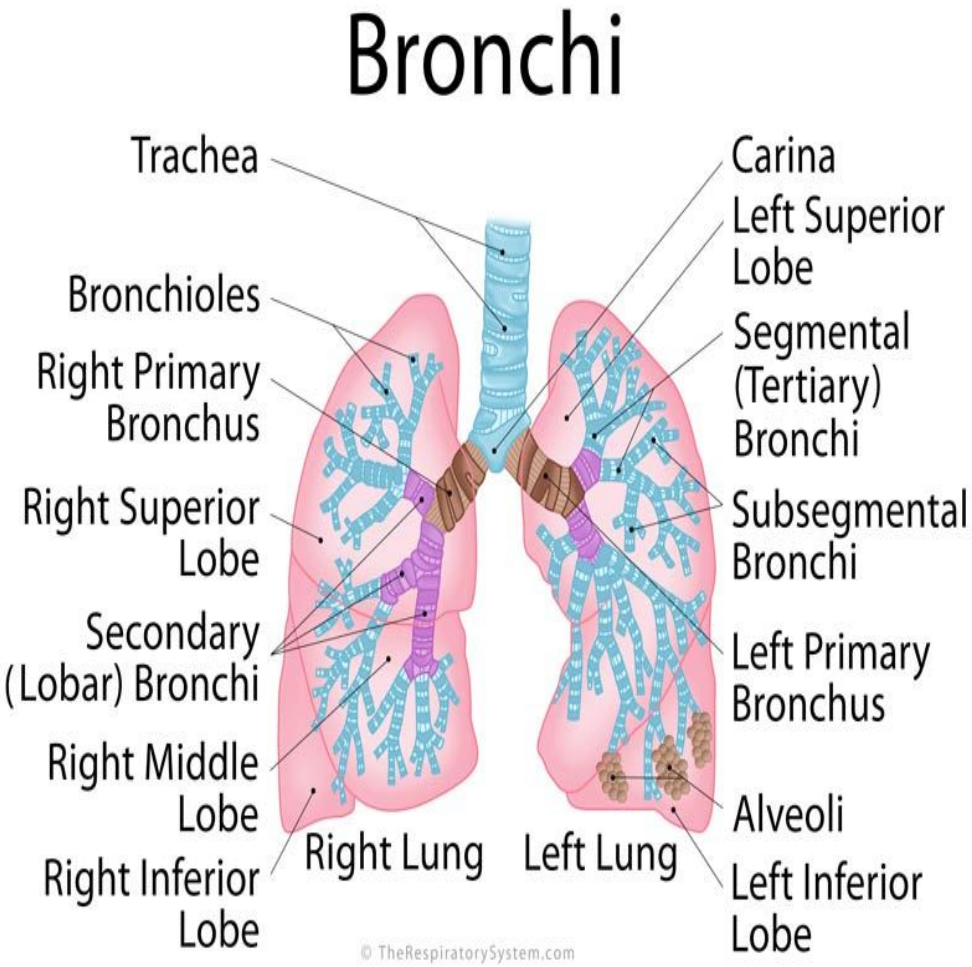
- Similar in structure to trachea
- Right bronchus is wider, shorter, and more vertical than left.
- Primary bronchi enter lung at hilum

Secondary Bronchi

- One bronchi per lobe of lung
- Three on right side
- Two on left side

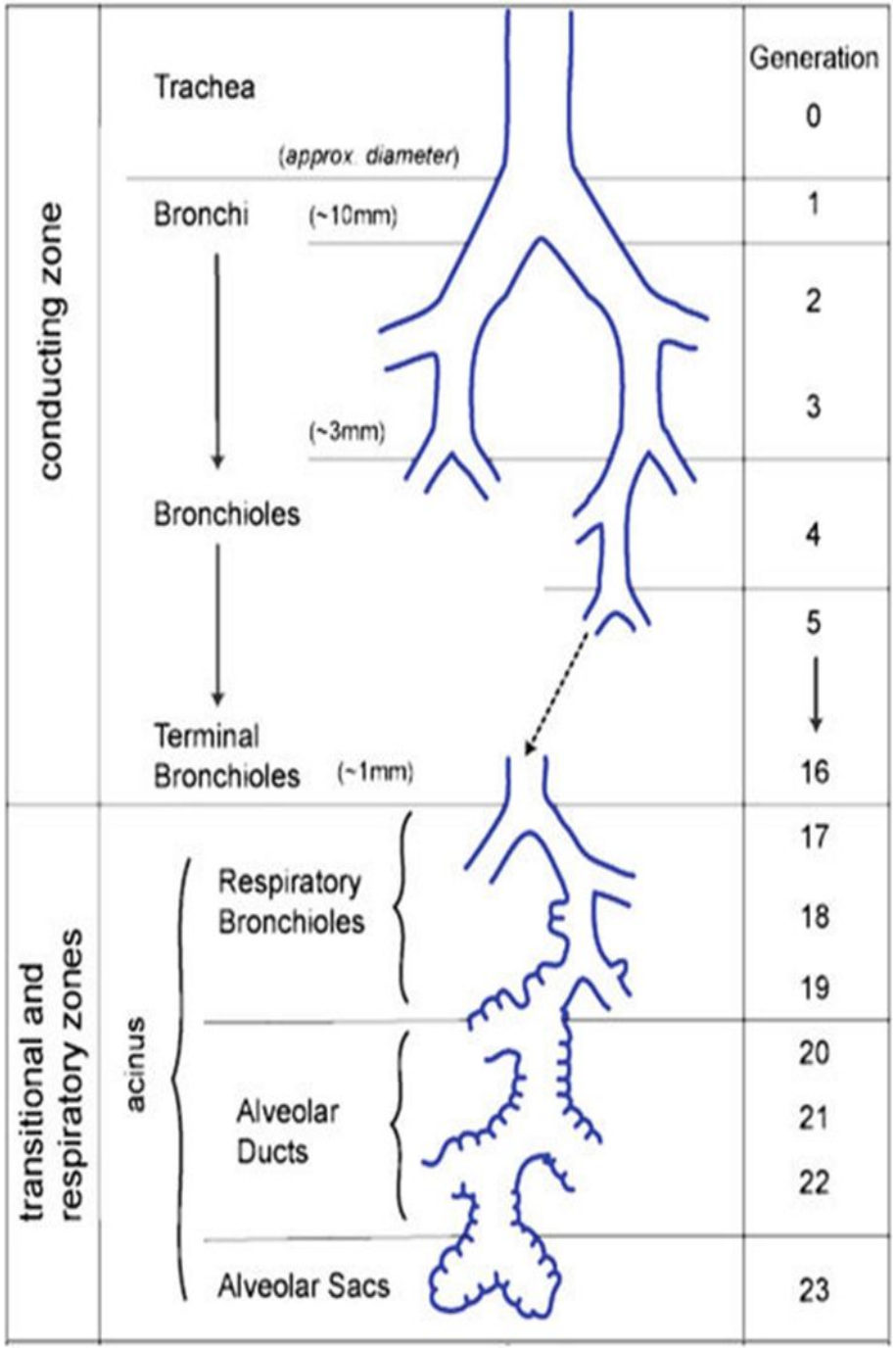
Tertiary Bronchi

- Tertiary bronchi supply bronchopulmonary segments
- Right lung has 10 segments
- Left lung has 8-9 segments
- Tertiary bronchus branch several times and eventually form bronchioles.



Generations of Bronchial Tree

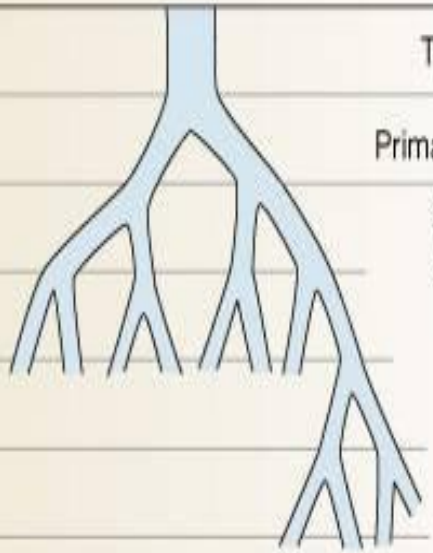
- Conducting Zone up to 16th generation or up to Terminal Bronchioles.
- Air not involved In gas exchange, called Dead Space Air.
- Normal Value of dead space air = 150ml.
- From 17th generation to 23rd or from respiratory bronchioles to alveolar sacs is gas exchange area.



RESPIRATORY ZONE

- **The respiratory zone** includes the structures that are lined with alveoli and, therefore, participate in gas exchange: the respiratory bronchioles, the alveolar ducts, and the alveolar sacs.
- The respiratory bronchioles are transitional structures. Like the conducting airways, they have cilia and smooth muscle, but they also are considered part of the gas exchange region because alveoli occasionally bud off their walls. The alveolar ducts are completely lined with alveoli, but they contain no cilia and little smooth muscle. The alveolar ducts terminate in alveolar sacs, which also are lined with alveoli.

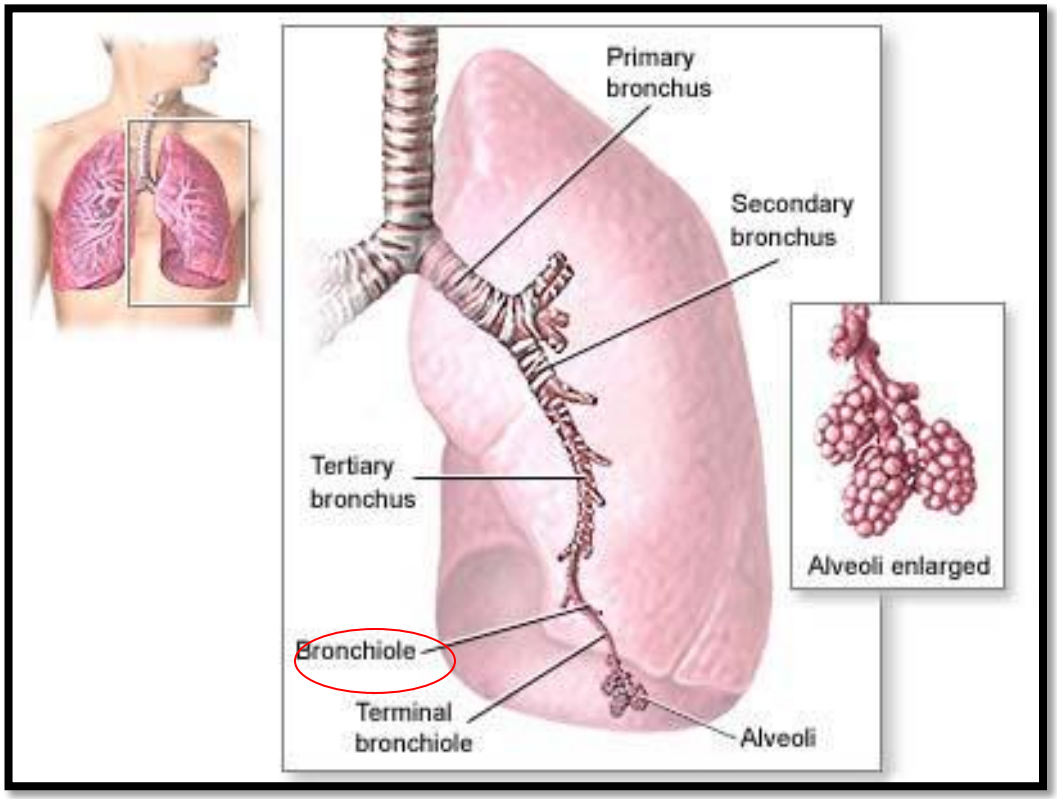
Comparison of conducting and respiratory zones



	Name	Division	Diameter (mm)	How many?	Cross-sectional area (cm)
Conducting system	Trachea	0	15-22	1	2.5
	Primary bronchi	1	10-15	2	↓
	Smaller bronchi	2	1-10	4	↓
		3		↓	↓
		4		↓	↓
		5		↓	↓
		6-11		1×10^4	↓
Exchange surface	Bronchioles	12-23	0.5-1	2×10^4 ↓ 8×10^7	100 ↓ 5×10^3
	Alveoli	24	0.3	$3\text{-}6 \times 10^8$	$>1 \times 10^6$

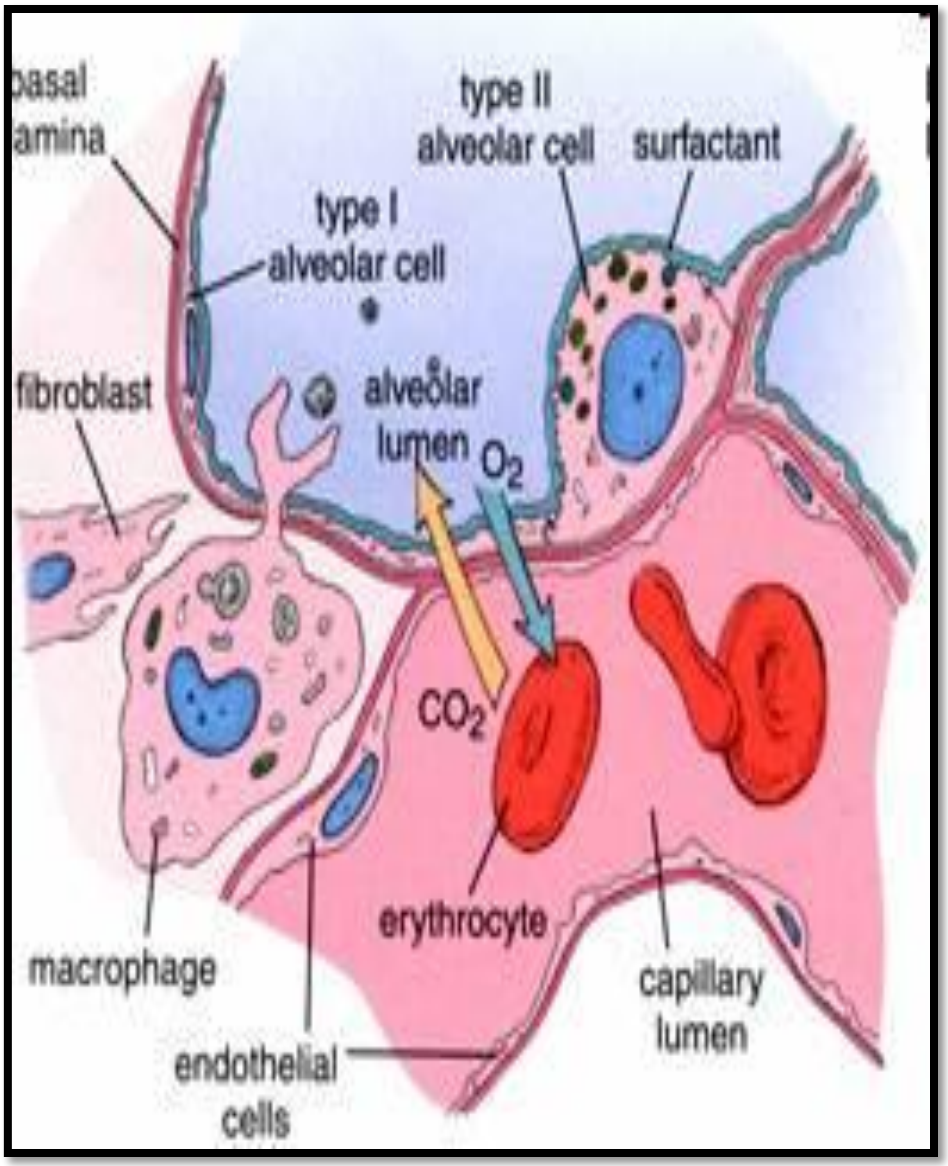
Bronchioles

- Respiratory passages less than 1 mm in diameter
- Branch several times into terminal bronchioles which extend into gas exchange bronchioles
- Alveoli can extend from respiratory bronchioles



Alveoli

- Each lung contains **150 million**
- **Type 1 pneumocytes = epithelium**
- **Type II pneumocytes = create surfactant**
- Macrophage - Phagocytize debris

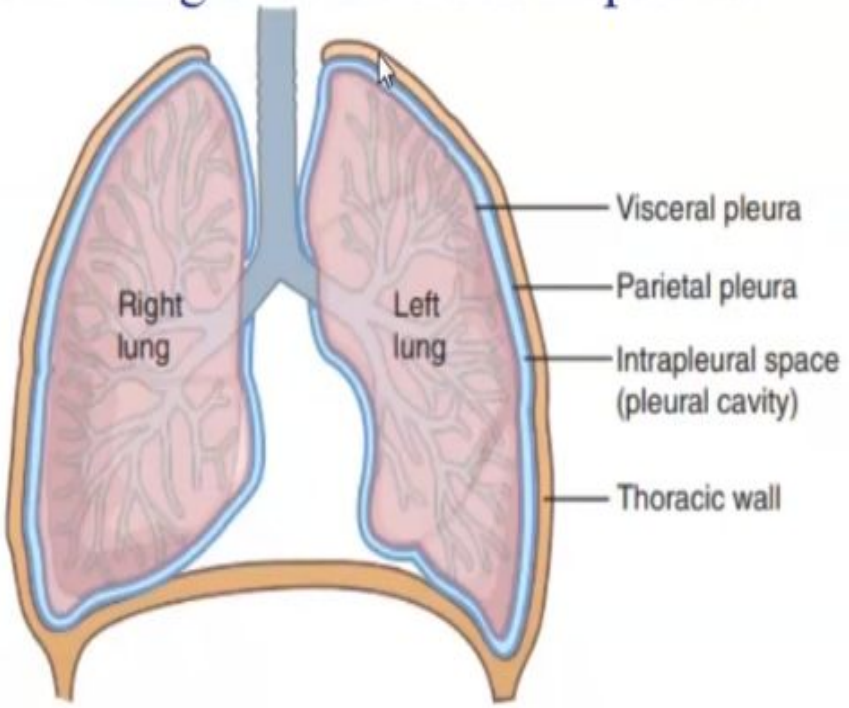


Lungs

- Covered by pleura
- Thin double layered serous membrane
- Secretes pleural fluid to lubricate lungs inside chest cavity
- Visceral layer covers lung
- Parietal layer covers the inside of chest cavity
- Pleural cavity; potential space between two pleural layers

PLEURA, PLEURAL CAVITY AND INTRA-PLEURAL PRESSURE

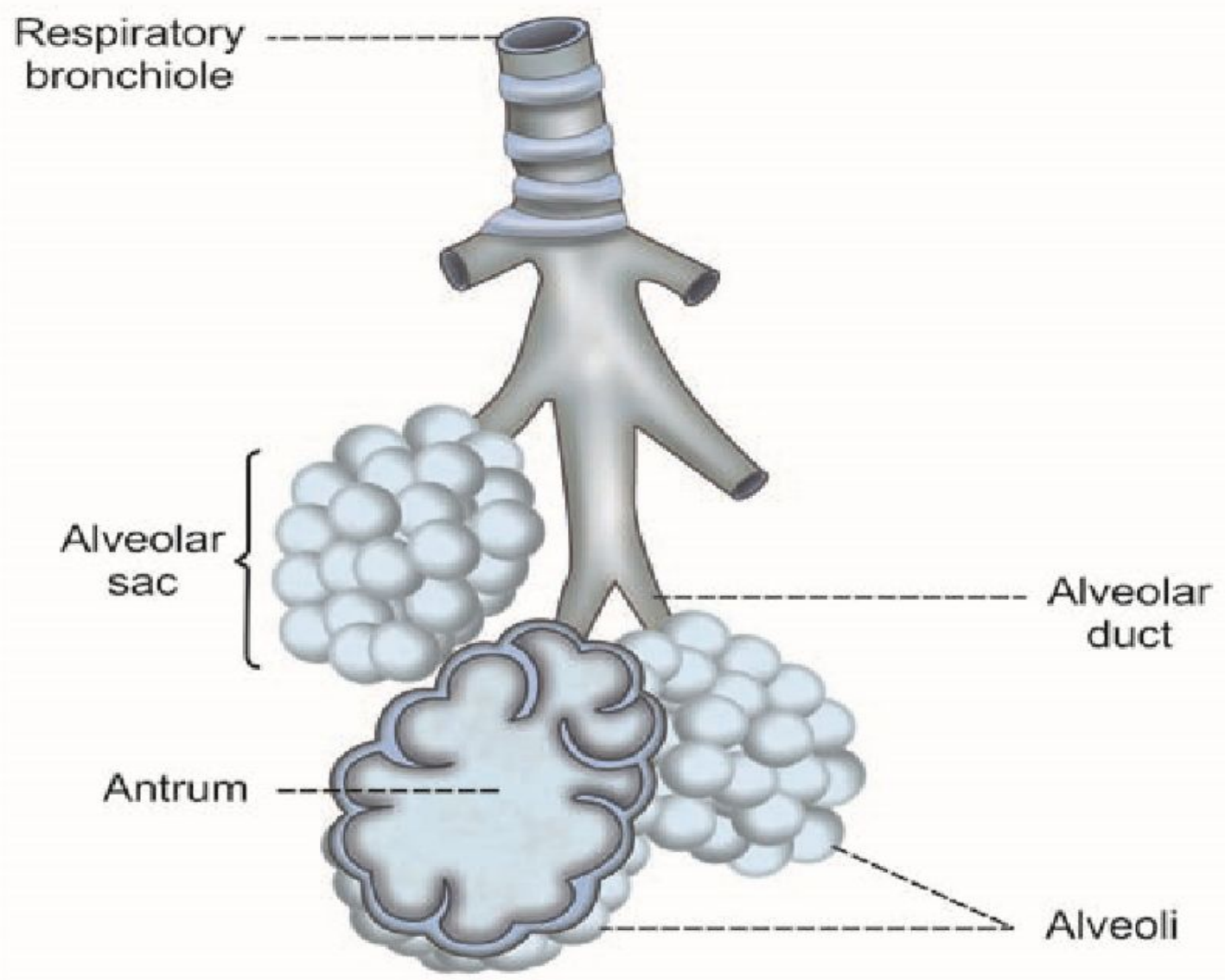
- The lungs are covered by visceral pleura which are reflected over the inner aspect of the chest wall enclosing a potential cavity between its two layers known as pleural cavity.
- The two layers, parietal and visceral are kept moist and lubricated by a few millilitres (2 ml) of a mucopolysaccharide containing fluid in the intra-pleural space so that during respiration the lungs with visceral pleura surrounding it glide smoothly over the parietal pleura.



Intra-pleural pressure:

- **Intrapleural pressure** refers to the pressure within the pleural cavity & is negative and amounts to -2.5 mm Hg at the end expiratory position.

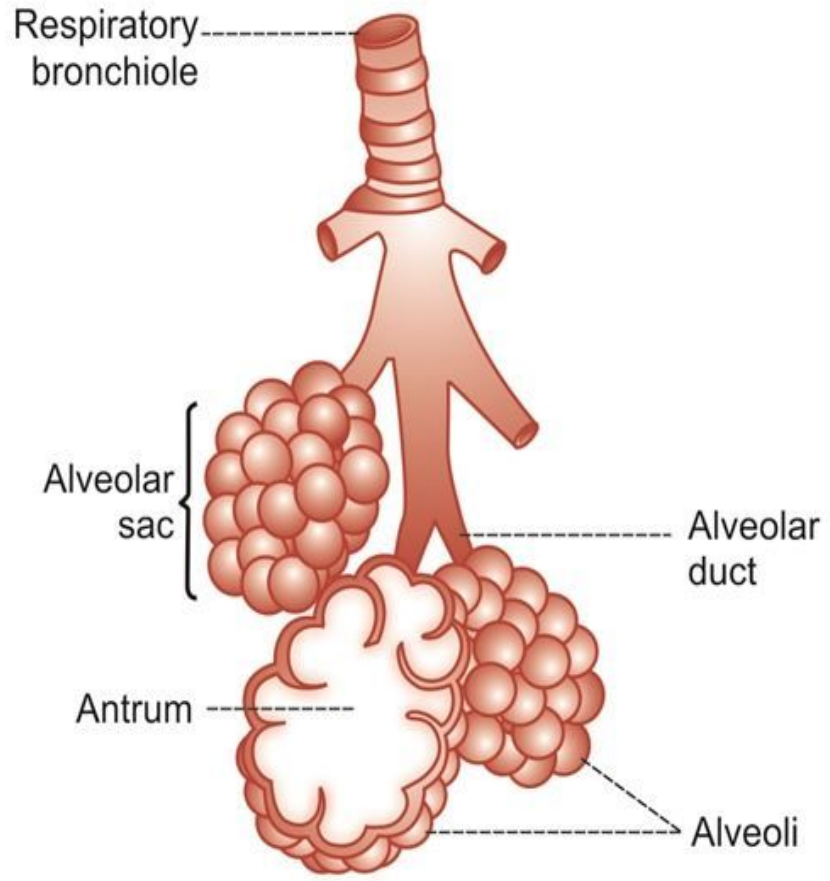
RESPIRATORY UNIT

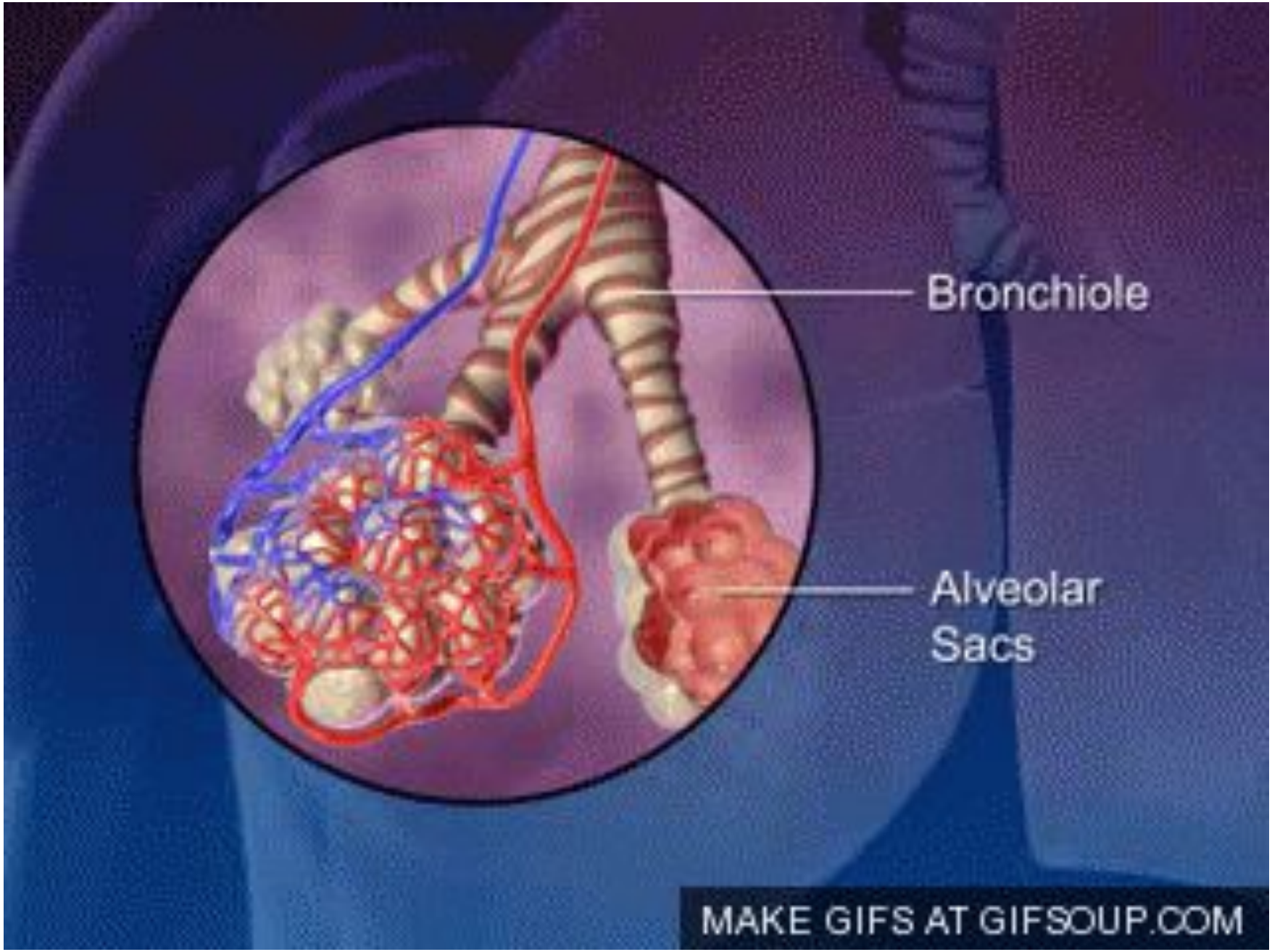


RESPIRATORY UNIT

- **Respiratory unit includes:**

1. Respiratory bronchioles
2. Alveolar ducts
3. Alveolar sacs
4. Antrum
5. Alveoli




Bronchiole
Alveolar
Sacs
MAKE GIFS AT GIFSOUP.COM

FUNCTIONS OF RESPIRATORY SYSTEM

- Functions of respiratory system including lungs can be divided into **Respiratory and non-Respiratory Functions**.

Respiratory Functions:

1. **Gas Exchange:** Lungs are the central structure in respiration. Inhalation of oxygen into the body and removal of carbon dioxide from the body take place through lungs.
2. **Regulation of blood pH:** By controlling carbon dioxide output from the body, lungs control plasma bicarbonate concentration. Therefore, lungs play an important role in acid-base balance.

FUNCTIONS OF RESPIRATORY SYSTEM

Non-Respiratory Functions:

1. **Olfaction:** Olfactory receptors present in the mucous membrane of nostrils are responsible for olfactory sensation.
2. **Vocalization:** Along with other structures, larynx form speech apparatus. However, larynx plays major role in the process of vocalization. Therefore, it is called sound box.
3. **Prevention of Dust Particles:** Dust particles, which enter the nostrils through air, are prevented from reaching the lungs by filtration action of the hair in nasal mucous membrane. Those dust particles, which escape nasal hair and nasal mucous membrane, are removed by the phagocytic action of macrophages in the alveoli.

FUNCTIONS OF RESPIRATORY SYSTEM

4. Defence Functions: Respiratory system is involved in defense functions in following ways:

i. Organisms that enter the lungs are phagocytosed by **pulmonary alveolar macrophages** (dust cells) or interstitial macrophages in the lungs.

ii. IgA present in bronchial secretions provides resistance to infection. IgA also maintains integrity of the mucosal lining of the respiratory tract.

5. Maintenance of Water Balance: Respiratory tract plays a major role in water loss mechanism. During expiration, water evaporates through the expired air and some amount of body water is lost by this process.

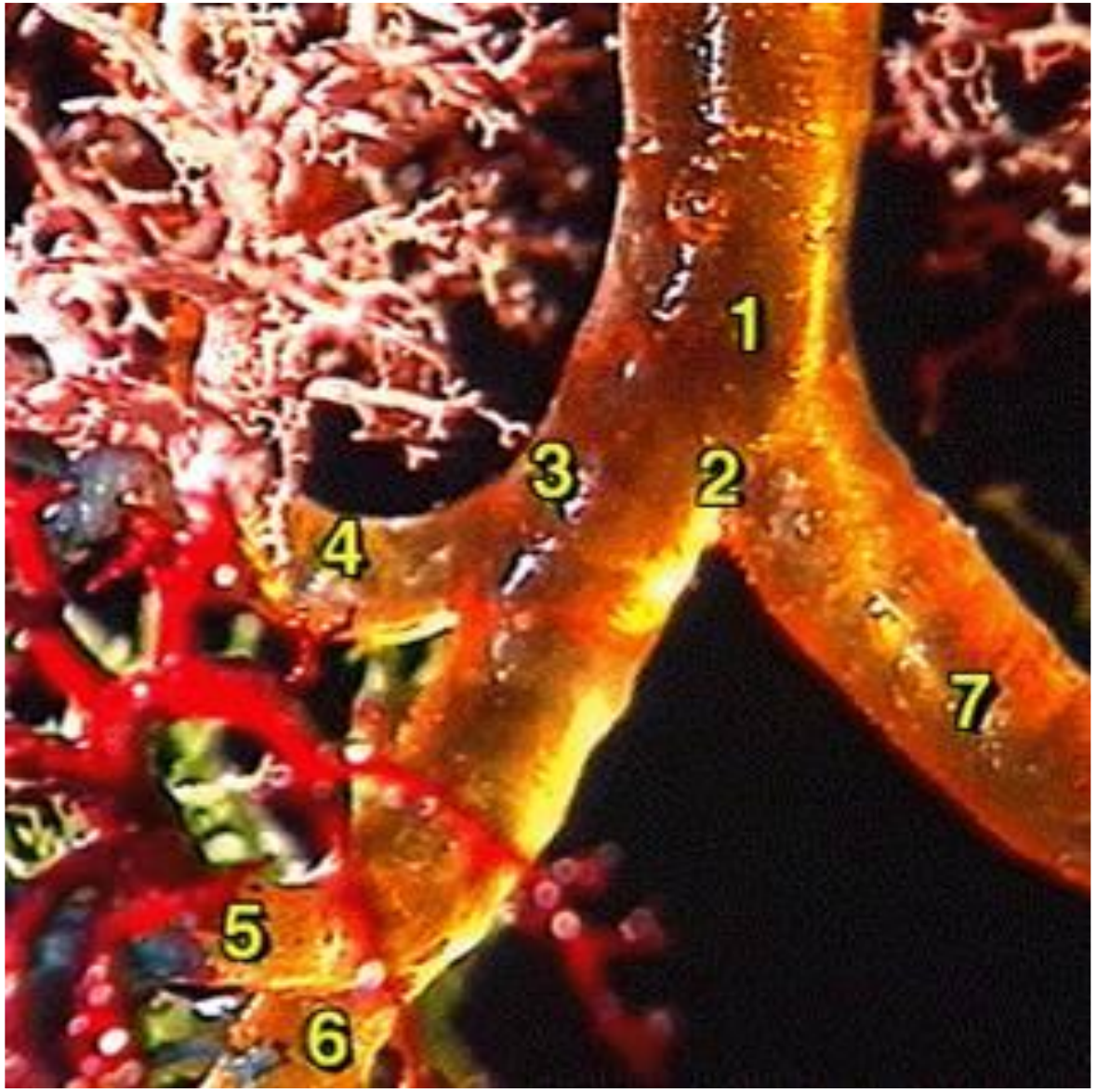
FUNCTIONS OF RESPIRATORY SYSTEMS

6. Regulation of body temperature: During expiration along with water heat is also lost through the body. Thus, respiratory tract plays a role in heat loss mechanism.

7. Anticoagulant Function: **Mast cells** in lungs secret **heparin**. Heparin is a anticoagulant and it prevents the intravascular clotting.

8. Endocrine Functions: Angiotension converting enzyme is mainly secreted by endothelium of pulmonary blood vessels that converts Angiotension I to Angiotension II, which occur in pulmonary secretion.

Identify the structures in sequence



Answers

1. Trachea
2. Carina
3. Right main bronchus
4. Right superior lobe bronchus
5. Right middle lobe bronchus
6. Right lower lobe bronchus
7. Left main bronchus

Identify the inspiration and expiration phase in both lungs

